

Distance-Based Diagnostics for Batch Effects

Technical Alignment and Biological Structure Preservation

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Goal of the Analysis

Question 1: Technical batch alignment

Are the same biological cell types close across batches?

$$C_t^{(b_1)} \text{ vs. } C_t^{(b_2)}$$

Example:

$$C_\alpha^{(\text{celseq2})} \text{ vs. } C_\alpha^{(\text{smartseq2})}$$

Question 2: Biological structure preservation

Inside each batch, is the relationship between alpha and other cell types preserved?

$$C_\alpha^{(b)} \text{ vs. } C_t^{(b)}$$

Notation I: Expression Matrix

For each batch b , let

$$X^{(b)} \in \mathbb{R}^{n_b \times p}$$

where

n_b = number of cells in batch b , p = number of genes.

Each row is one cell:

$$\mathbf{x}_i^{(b)} = (x_{i1}^{(b)}, x_{i2}^{(b)}, \dots, x_{ip}^{(b)}) \in \mathbb{R}^p.$$

Interpretation

Each cell is treated as a point in p -dimensional gene-expression space.

Notation II: Cell-Type Sets

For a cell type t , define

$$C_t^{(b)} = \{i : \text{cell } i \text{ in batch } b \text{ has type } t\}.$$

Examples:

$$C_\alpha^{(b)} = \{i : \text{cell } i \text{ is alpha in batch } b\},$$

$$C_\beta^{(b)} = \{i : \text{cell } i \text{ is beta in batch } b\}.$$

Main object

We compare distances between sets of cells, not just individual cells.

Notation III: Euclidean Distance

For two cells i and j , define

$$d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2.$$

Expanded over genes:

$$d(i, j) = \sqrt{\sum_{g=1}^p (x_{ig} - x_{jg})^2}.$$

Meaning

A larger distance means the two cells have more different expression profiles.

Methods Snapshot

- Dataset: five pancreatic single-cell batches:

`celseq`, `celseq2`, `fluidigm1`, `indrop`, `smartseq2`.

- Distances are computed in gene-expression space:

$$\mathbf{x}_i \in \mathbb{R}^p.$$

- Main diagnostics:

- 1 Matching cell-type distances across batches.
- 2 Alpha-centered relative distances within batches.

- Rare cell-type comparisons are treated cautiously because pairwise distances become unstable with small cell counts.

Dataset Composition

Cell type	celseq	celseq2	fluidigm1	indrop	smartseq2
acinar	229	274	21	1152	188
activated stellate	19	90	16	294	55
alpha	213	844	241	2309	1008
beta	161	445	258	2507	308
delta	50	203	25	608	127
ductal	304	257	34	915	444
endothelial	5	21	14	235	21
gamma	18	110	18	266	213

Why this matters

Major cell types have enough cells for stable distance distributions. Rare cell types are excluded or interpreted cautiously.

Analysis 1: Matching Cell-Type Distances

For each batch pair (b_1, b_2) and cell type t , compute

$$\mathcal{D}_t^{(b_1, b_2)} = \left\{ d(i, j) : i \in C_t^{(b_1)}, j \in C_t^{(b_2)} \right\}.$$

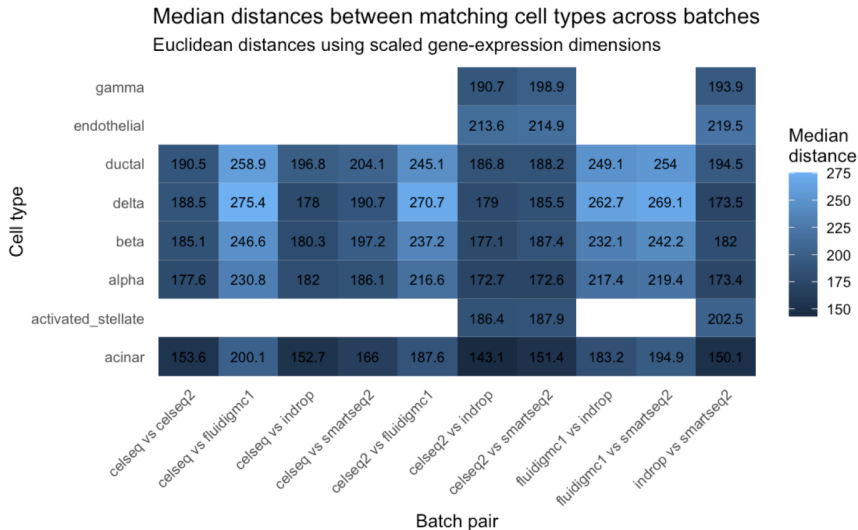
Then summarize the distribution by

$$M_t^{(b_1, b_2)} = \text{median} \left(\mathcal{D}_t^{(b_1, b_2)} \right).$$

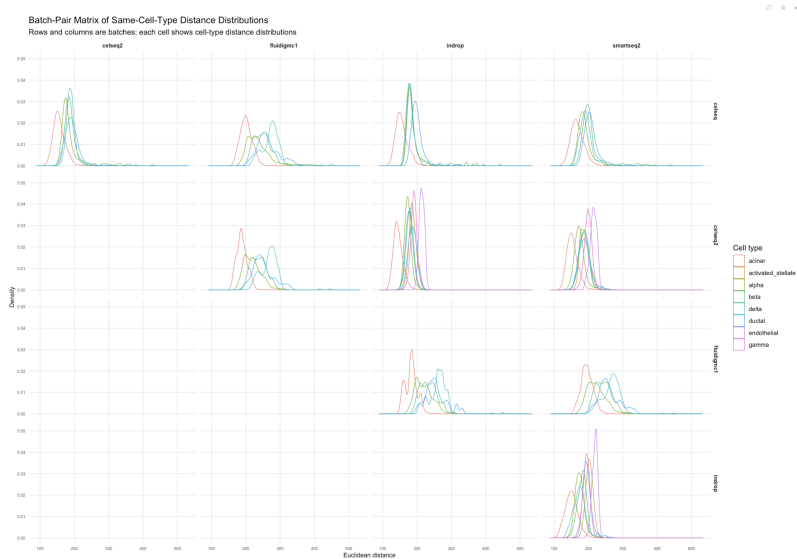
Interpretation

Large $M_t^{(b_1, b_2)}$ means the same cell type is far apart across the two batches.

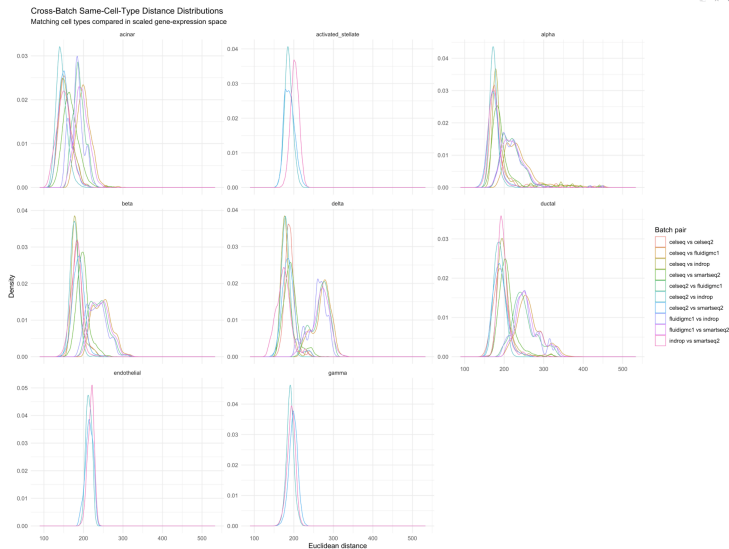
Plot 1: Matching Cell-Type Median Distances



Plot 2: Pairwise Batch Comparison of Cell-Type Distance Distributions



Plot 3: Same-Cell-Type Distance Distributions Across Batch Pairs



Reading Plots 1–3: Technical Batch Alignment

- **Plot 1: Median heatmap** gives one summary value for each cell type and batch pair:

$$M_t^{(b_1, b_2)} = \text{median} \left(\mathcal{D}_t^{(b_1, b_2)} \right).$$

This is the clearest ranking of which batch pairs are most separated.

- **Plot 2: Batch-pair matrix** shows, for each pair of batches, the distance distributions of several cell types together. This helps identify whether a batch pair is globally shifted, or whether only some cell types are affected.
- **Plot 3: Cell-type-wise distributions** fixes the cell type and compares all batch pairs. This shows which cell types are stable across technologies and which are more batch-sensitive.

Main reading

Plot 1 gives the ranking, Plot 2 shows batch-pair behavior, and Plot 3 shows cell-type-specific behavior.

Result 1: Technical Batch Alignment

- fluidigm1 is the clearest technical outlier:

$M_t^{(\text{fluidigm1}, b)}$ is often large across several t .

- The strongest separation is not uniform across cell types. Endocrine and ductal populations show stronger shifts than some other groups:

$M_{\alpha}^{(b_1, b_2)}$, $M_{\beta}^{(b_1, b_2)}$, $M_{\delta}^{(b_1, b_2)}$, $M_{\text{ductal}}^{(b_1, b_2)}$ behave differently.

- celseq2, indrop, and smartseq2 appear more mutually compatible.

Conclusion

The first diagnostic shows that batch effects are both **batch-specific** and **cell-type-specific**, with fluidigm1 driving the strongest technical separation.

Analysis 2: Alpha-Centered Geometry

Now fix alpha cells as an anchor population.

Inside each batch b , compute the alpha-to-alpha distance set:

$$\mathcal{A}^{(b)} = \left\{ d(i, j) : i, j \in C_{\alpha}^{(b)}, i < j \right\}.$$

Define the alpha baseline:

$$B_{\alpha}^{(b)} = \text{median} \left(\mathcal{A}^{(b)} \right).$$

Interpretation

$B_{\alpha}^{(b)}$ is the typical within-alpha distance in batch b .

Relative Alpha-to-Cell-Type Distance

For cell type t , define the relative distance

$$r_{ij}^{(b,t)} = \frac{d(i,j)}{B_{\alpha}^{(b)}}, \quad i \in C_{\alpha}^{(b)}, \quad j \in C_t^{(b)}.$$

That is,

$$r_{ij}^{(b,t)} = \frac{\|\mathbf{x}_i^{(b)} - \mathbf{x}_j^{(b)}\|_2}{\text{median} \left\{ d(u,v) : u, v \in C_{\alpha}^{(b)}, u < v \right\}}.$$

How to Read Relative Distance

$$r_{ij}^{(b,t)} \approx 1$$

means alpha-to- t distance is similar to normal alpha-to-alpha variation.

$$r_{ij}^{(b,t)} > 1$$

means cell type t is farther from alpha than alpha cells are from each other.

$$r_{ij}^{(b,t)} < 1$$

means cell type t is unusually close to alpha relative to alpha variability.

Why Relative Distance Matters

Raw distance is affected by the batch-specific scale:

$$d(i, j) = \text{global batch scale} + \text{biological separation}.$$

Relative distance rescales by within-alpha spread:

$$r_{ij}^{(b,t)} = \frac{d(\alpha, t)}{\text{median } d(\alpha, \alpha)}.$$

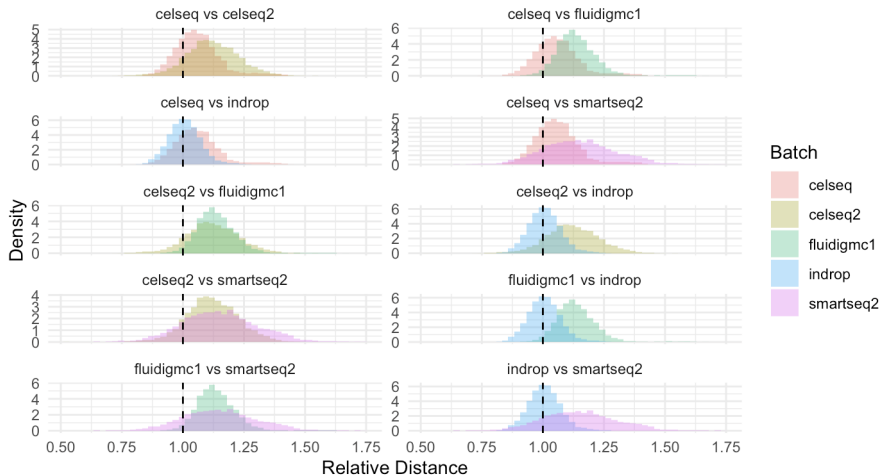
Key idea

Raw distance measures absolute magnitude. Relative distance measures local biological geometry around alpha cells.

Plot 4: Relative Alpha-to-Beta Distances

Relative alpha-to-beta distance distributions

All two-batch combinations using normalized original gene dimensions



Interpreting Plot 4: Relative Alpha-to-Beta Distances

- The dashed line at 1 represents the within-batch alpha-to-alpha baseline. Values to the right of this line mean beta cells are farther from alpha than alpha cells are from each other.
- Most distributions are shifted to the right of 1, so alpha and beta cells are generally separated in all batches.
- The largest discrepancy appears between `indrop` and `smartseq2`. `indrop` is often closer to the baseline, while `smartseq2` is more right-shifted and broader. This suggests stronger alpha-beta separation in `smartseq2`.
- Looking at one batch as an anchor, such as `celseq`, shows that different batches distort the alpha-beta distribution differently. For example, comparing `celseq` against `celseq2`, `fluidigm1`, `indrop`, and `smartseq2` gives different shifts and shapes.
- Therefore, the batch effect is not just changing the average distance; it is also changing the shape and spread of the biological relationship.

Main insight

Alpha-beta structure is broadly present across batches, but the strength and shape of the separation vary by technology.

Mean Relative Separation

For each batch b and cell type t , define

$$\hat{\mu}_{\alpha,t}^{(b)} = \frac{1}{|C_{\alpha}^{(b)}| |C_t^{(b)}|} \sum_{i \in C_{\alpha}^{(b)}} \sum_{j \in C_t^{(b)}} r_{ij}^{(b,t)}.$$

Meaning

$\hat{\mu}_{\alpha,t}^{(b)}$ measures the average relative separation between alpha and cell type t inside batch b .

Signed Difference Between Batches

For two batches b_1 and b_2 , define

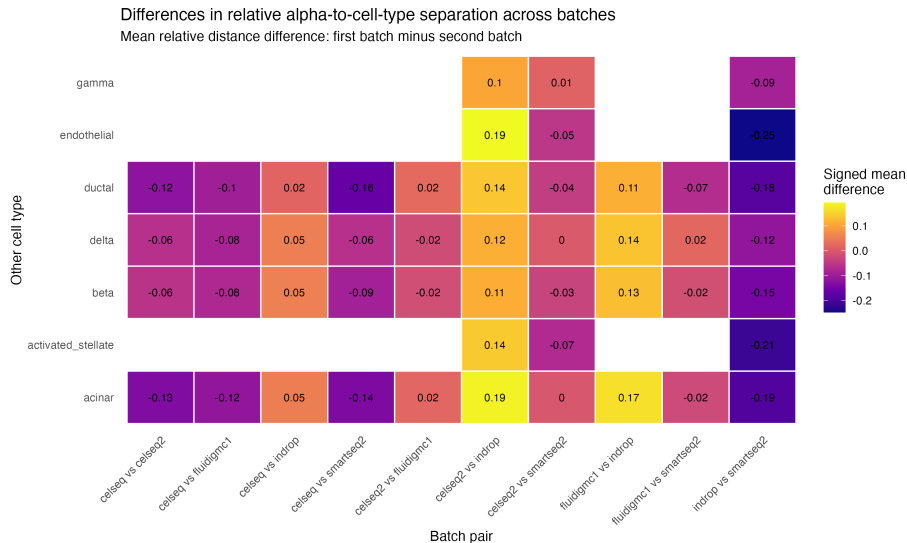
$$\Delta_{\alpha,t}^{(b_1,b_2)} = \hat{\mu}_{\alpha,t}^{(b_1)} - \hat{\mu}_{\alpha,t}^{(b_2)}.$$

$$\Delta_{\alpha,t}^{(b_1,b_2)} > 0 \quad \Rightarrow \quad t \text{ is farther from alpha in } b_1.$$

$$\Delta_{\alpha,t}^{(b_1,b_2)} < 0 \quad \Rightarrow \quad t \text{ is farther from alpha in } b_2.$$

$$\Delta_{\alpha,t}^{(b_1,b_2)} \approx 0 \quad \Rightarrow \quad \text{similar alpha-to-}t \text{ geometry.}$$

Plot 5: Signed Relative Geometry Difference



Result 2: Biological Geometry Around Alpha

- `celseq2` and `smartseq2` have small signed differences for many cell types:

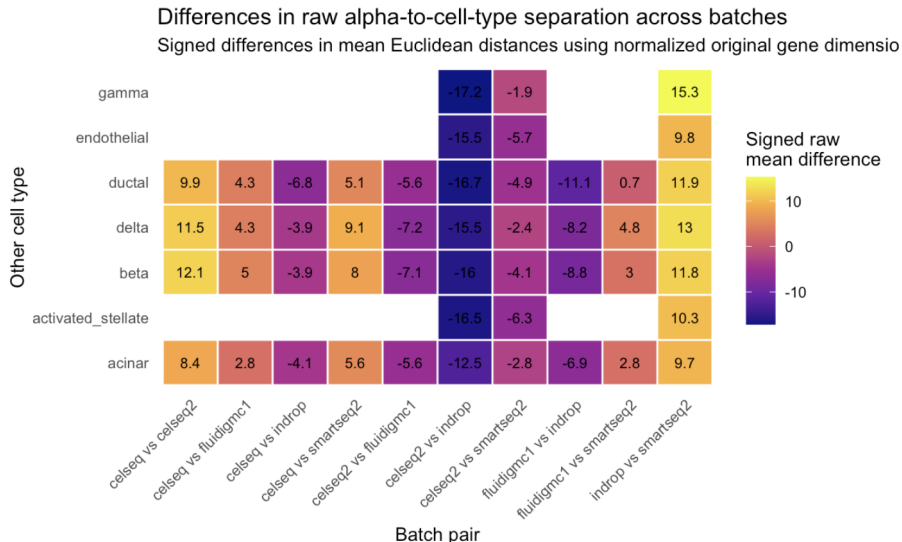
$$\Delta_{\alpha,t}^{(\text{celseq2}, \text{smartseq2})} \approx 0.$$

- `indrop` vs `smartseq2` is often negative:

$$\Delta_{\alpha,t}^{(\text{indrop}, \text{smartseq2})} < 0.$$

- This suggests `smartseq2` often shows stronger alpha-to-other separation than `indrop`.

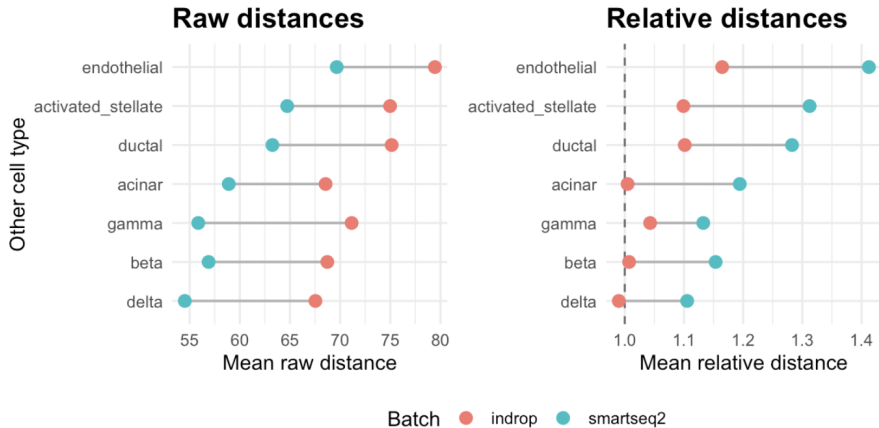
Plot 6: Signed Raw Geometry Difference



Plot 7: Raw Versus Relative Example

Raw versus relative alpha-to-cell-type distances

inDrop versus Smart-seq2 using original gene-expression dimensions



Interpretation: Raw vs Relative Distances

The raw panel shows that `indrop` often has larger absolute alpha-to-cell-type distances:

$$d_{\text{raw}}^{(\text{indrop})}(\alpha, t) > d_{\text{raw}}^{(\text{smartseq2})}(\alpha, t).$$

However, the relative panel shows the opposite pattern for many cell types:

$$r^{(\text{smartseq2}, t)} > r^{(\text{indrop}, t)}.$$

Meaning

`indrop` has larger absolute distances, but it also has larger within-alpha spread. After normalizing by alpha-to-alpha distance, `smartseq2` shows stronger alpha-to-other-cell-type separation.

Why the Reversal Matters

Raw distance answers:

How large are the distances in absolute gene-expression space?

Relative distance answers:

How far is a cell type from alpha compared with alpha's own variability?

Key interpretation

The reversal suggests that the difference is not only biological separation. Part of the raw distance difference comes from batch-specific scale.

- indrop: larger raw distances, but weaker relative alpha-to-other separation.
- smartseq2: smaller raw distances, but stronger relative biological separation.

Raw vs Relative: Agreement and Disagreement

Raw distance	Relative distance	Interpretation
High	High	Large absolute scale + strong biological separation
High	Low	Global spread is large, but biology is compressed
Low	High	Small absolute scale, but strong biological separation
Low	Low	Small scale + weak biological separation

Key idea

Raw distance measures absolute magnitude:

$$d(\alpha, t)$$

Relative distance measures separation after normalizing by alpha variability:

$$r(\alpha, t) = \frac{d(\alpha, t)}{\text{median } d(\alpha, \alpha)}.$$

Reconciling the Two Analyses: Three Signals

The diagnostics measure three different signals:

Signal	Question answered	Interpretation
Matching-cell distance	Are same cell types close across batches?	Technical alignment
Raw alpha distance	Are absolute within-batch distances large?	Global scale effect
Relative alpha distance	Is alpha-to-cell-type geometry preserved?	Biological structure

Key point

These are not interchangeable. A batch can look aligned in one diagnostic but distorted in another.

Reconciling the Two Analyses: Main Logic

- **Matching-cell-type distances** compare the same cell type across batches:

$$\alpha^{(b_1)} \text{ vs. } \alpha^{(b_2)}.$$

This tells us whether batches are technically aligned.

- **Alpha-centered distances** compare alpha to other cell types inside each batch:

$$\alpha^{(b)} \text{ vs. } \beta^{(b)}, \quad \alpha^{(b)} \text{ vs. } \delta^{(b)}, \quad \alpha^{(b)} \text{ vs. ductal}^{(b)}.$$

This tells us whether biological geometry is preserved.

- Therefore, batch effects can be decomposed conceptually as:

$$\text{Batch effect} = \text{cross-batch shift} + \text{within-batch geometry distortion}.$$

Combined Interpretation Matrix

Same-cell distance	Relative geometry difference	Meaning
Low	Low	Best case: batches align and biological structure is preserved.
High	Low	Global batch shift: same cell types are far across batches, but internal cell-type geometry is preserved.
Low	High	Hidden distortion: batches may look aligned, but the biological relationship between cell types changes.
High	High	Worst case: technical batch shift and biological-structure distortion occur together.

How to use this matrix

Matching-cell distances tell us whether batches are technically separated. Relative geometry tells us whether biological structure is preserved or distorted.

Key Numerical Findings

1. Strongest technical separation

Pairs involving `fluidigm1` have the largest matching-cell-type medians.

Cell type	Batch pair	Median distance
δ	<code>celseq</code> vs <code>fluidigm1</code>	275.4
δ	<code>celseq2</code> vs <code>fluidigm1</code>	270.7
δ	<code>fluidigm1</code> vs <code>smartseq2</code>	269.1
ductal	<code>celseq</code> vs <code>fluidigm1</code>	258.9
β	<code>celseq</code> vs <code>fluidigm1</code>	246.6

Interpretation: `fluidigm1` is the clearest technical outlier.

2. Most compatible batch group

The lower-distance group is approximately

`celseq2`, `indrop`, `smartseq2`.

For major cell types, the best average median distances are roughly:

Batch pair	Average median
<code>celseq2</code> vs <code>indrop</code>	171.7
<code>indrop</code> vs <code>smartseq2</code>	174.7
<code>celseq2</code> vs <code>smartseq2</code>	177.0

Interpretation: these batches are more mutually aligned than pairs involving `fluidigm1`.

Key Numerical Findings: Alpha-Centered Geometry

3. Stable alpha-centered geometry

celseq2 vs smartseq2 has small signed differences:

Cell type	$\Delta_{\alpha,t}^{(\text{celseq2}, \text{smartseq2})}$
acinar	0.00
β	-0.03
δ	0.00
ductal	-0.04
γ	0.01

Interpretation: alpha-centered geometry is relatively similar between these two batches.

4. Strongest relative-geometry discrepancy

indrop vs smartseq2 is strongly negative:

Cell type	$\Delta_{\alpha,t}^{(\text{indrop}, \text{smartseq2})}$
endothelial	-0.25
activated stellate	-0.21
acinar	-0.19
ductal	-0.18
β	-0.15
δ	-0.12

Interpretation: smartseq2 shows stronger relative alpha-to-other separation than indrop.

Main Conclusions

- ① **fluidigm1 is the strongest technical outlier.** Same-cell-type cross-batch distances are consistently larger when this batch is involved.
- ② **celseq2 and smartseq2 are the most structurally stable pair.** They show relatively low matching-cell-type distances and similar alpha-centered geometry.
- ③ **indrop shows a mismatch between cross-batch alignment and internal geometry.** It can appear reasonably compatible in matching-cell-type cross-batch distances, but its alpha-centered relative distances suggest compressed biological structure around alpha cells.
- ④ **Batch effects are not uniform.** They vary by both technology pair and cell type, so a single global batch-effect score would hide important cell-type-specific behavior.

Overall conclusion

We need both diagnostics:

technical alignment + biological structure preservation.

Suggested Next Step

Run the same diagnostics before and after correction.

A good correction should reduce same-cell-type cross-batch distances:

$$\tilde{M}_t^{(b_1, b_2)} < M_t^{(b_1, b_2)}.$$

It should also reduce disagreement in relative geometry:

$$\left| \tilde{\Delta}_{\alpha, t}^{(b_1, b_2)} \right| < \left| \Delta_{\alpha, t}^{(b_1, b_2)} \right|, \quad \tilde{\Delta}_{\alpha, t}^{(b_1, b_2)} \approx 0.$$

Correction target

Align matching cell types across batches while keeping different cell types biologically separated.

Batch correction should **not only** make
batches closer.

It should make **matching cell types** closer
while preserving **biological geometry**.